

# The Observer World-Tube Principle: A Variational Extension of the Quantum Factorization Problem

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## Abstract

Tegmark proposed that observer-like physical systems may be distinguished by information, integration, independence, and nontrivial dynamics, and connected these properties to the problem of selecting a preferred tensor factorization of Hilbert space. A fixed factorization, however, identifies an observer with a fixed collection of degrees of freedom. Biological and computational identity can persist while physical components and internal representations change. We therefore propose that the relevant mathematical object is not a factorization but a trajectory through factorization space, called an observer world-tube. We introduce an exact Gaussian baseline that combines directed internal integration, conditional environmental insulation, and predictive persistence. We prove that the directed term rejects contemporaneously correlated but dynamically uncoupled systems, a failure mode of static mutual information. We then formulate a regularized action for time-dependent subsystem boundaries and recover a planted moving module in an initial numerical experiment. The score is not presented as a measure of phenomenal consciousness. It is a falsifiable candidate principle for selecting persistent observer-like organization, and a route toward a gauge-invariant quantum formulation.

## 1 Motivation

Tegmark’s perceptronium program asks which physical properties distinguish observer-like matter and whether those properties can help select the spatial factorization perceived by observers [4]. The program exposes a tension. Independence favors a weak interaction Hamiltonian, but maximizing it alone selects dynamically trivial structures. Conscious organization, if it is physical, must be sufficiently insulated to have its own dynamics and sufficiently coupled internally to change as a whole.

Existing work already supplies temporal and causal measures of integration [2, 3] and transformation-invariant measures of dynamical independence [1]. Our proposal is therefore not merely to add a temporal mutual information term. We instead let the factorization itself change. This shifts the object of study from an instantaneous subsystem to a persistent path in a space of possible subsystems.

## 2 Exact Gaussian baseline

Let

$$X_{t+1} = AX_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, Q), \quad \rho(A) < 1. \quad (1)$$

The stationary covariance  $\Sigma$  uniquely solves

$$\Sigma = A\Sigma A^\top + Q. \quad (2)$$

All information quantities below are therefore analytic functions of  $A$  and  $Q$ .

For candidate subsystem  $S$  and bipartition  $S = U \sqcup V$ , define

$$J_\tau(U, V) = \mathcal{I}(X_U^{t+\tau}; X_V^t \mid X_U^t) + \mathcal{I}(X_V^{t+\tau}; X_U^t \mid X_V^t). \quad (3)$$

**Definition 1** (Directed integration).

$$\mathcal{J}_\tau(S) = \frac{1}{|S|} \min_{U \sqcup V = S} J_\tau(U, V). \quad (4)$$

The normalization is provisional and will be included in the ablation study. The minimum implements a weakest-link requirement. Unlike simultaneous mutual information, each cross-prediction conditions on the target part's own present.

**Definition 2** (Environmental leakage).

$$\mathcal{L}_\tau(S) = \frac{1}{|S|} \mathcal{I}(X_S^{t+\tau}; X_S^t \mid X_S^t). \quad (5)$$

Let  $P_\tau(S)$  be the mean squared canonical correlation between  $X_S^t$  and  $X_S^{t+\tau}$ . We define bounded factors

$$G_\tau = 1 - 2^{-\mathcal{J}_\tau}, \quad K_\tau = 2^{-\mathcal{L}_\tau}, \quad (6)$$

and a baseline score

$$\Omega_\tau(S) = (G_\tau K_\tau P_\tau)^{1/3}. \quad (7)$$

The geometric mean makes all three properties necessary. Other aggregations must be compared rather than assumed.

**Proposition 1** (Rejection of static common correlation). *Suppose  $A$  is diagonal and the innovations are independent across time, while  $Q$  may contain arbitrary contemporaneous cross-correlations. Then for every bipartition and every positive lag,  $J_\tau(U, V) = 0$ .*

*Proof.* For diagonal  $A$ ,  $X_U^{t+\tau}$  is a function of  $X_U^t$  and innovations after time  $t$ . Those innovations are independent of  $X_V^t$ . Hence  $X_U^{t+\tau} \perp X_V^t \mid X_U^t$ , so the first conditional mutual information vanishes. Interchanging  $U$  and  $V$  proves the second term also vanishes.  $\square$

This proposition does not solve causal identification in the presence of hidden variables. It establishes a narrower and useful guarantee: simultaneous correlation alone is insufficient.

### 3 Observer world-tubes

Let  $\mathfrak{F}_{d,n}$  be a space of factorizations  $\mathcal{H} \cong \mathcal{H}_S \otimes \mathcal{H}_E$  with dimensions  $d$  and  $n/d$ . An observer candidate is a curve

$$\gamma : t \mapsto F_t \in \mathfrak{F}_{d,n}. \quad (8)$$

In a finite classical system, its discrete analog is a sequence  $S_0, S_1, \dots, S_T$ . Its support in index-time space is

$$\mathcal{W} = \{(i, t) : i \in S_t\}, \quad (9)$$

which we call an observer world-tube.

We propose the action

$$\mathcal{A}(\mathcal{W}) = \sum_t [\alpha \mathcal{J}_\tau(S_t, S_{t+\tau}) - \beta \mathcal{L}_\tau(S_t, S_{t+\tau}) + \chi \mathcal{P}_\tau(S_t, S_{t+\tau})] \quad (10)$$

$$- \lambda \sum_t d_{\mathfrak{F}}(S_t, S_{t+1}). \quad (11)$$

The transport term  $\mathcal{P}$  rewards continuity of organization. The metric term penalizes arbitrary changes of factorization without requiring unchanging material membership.

For the quantum theory, tensor factorizations can be represented by unitary transformations modulo local basis changes. Schematically,

$$\mathfrak{F}_{d,n} \simeq U(n) / (U(d) \otimes U(n/d)), \quad (12)$$

subject to the correct stabilizer and discrete identifications. If  $U_t$  represents the factorization path, the horizontal part of  $U_t^\dagger \dot{U}_t$  measures genuine factorization drift, while the vertical part is a local relabeling. This is the point at which a gauge connection enters naturally.

**Conjecture 1** (Observer world-tube principle). *For physical systems containing a dynamically integrated and conditionally insulated process, a gauge-invariant version of  $\mathcal{A}$  has a stable local maximum whose world-tube follows organizational identity through changes in physical realization.*

## 4 Initial numerical experiment

We constructed a seven-node Gaussian system with a planted three-node coherent module. At each of five time steps, one node leaves and another enters. The implemented dynamic program evaluates all  $\binom{7}{3} = 35$  candidate boundaries at every step, combines the fixed-boundary baseline with directed transport and a boundary-motion cost, and exactly recovers the planted sequence

$$(0, 1, 2) \rightarrow (1, 2, 3) \rightarrow (2, 3, 4) \rightarrow (3, 4, 5) \rightarrow (4, 5, 6). \quad (13)$$

## 5 Falsification program

The proposal should be rejected or revised if any of the following persist under preregistered tests: the path is unstable under small coordinate changes; null systems produce equally stable world-tubes; fixed-factorization baselines perform equally well on moving-identity tasks; or the quantum action retains the same dynamically trivial optimum that motivated the extension.

The next mathematical tasks are to prove invariance properties, derive recovery bounds for planted moving modules, formalize the quotient geometry, and compare the action against geometric integrated information,  $\Phi$ ID, dynamical independence, causal emergence, and dynamic community detection.

## 6 Interpretation

The intended synthesis can be stated compactly. Space corresponds to a factorization that localizes interaction. Time is the ordering along which predictive structure changes. An observer is not a point in either space, but a persistent integrated path through both. The mathematics developed here does not explain subjective experience. It proposes a sharper physical question: which evolving boundary makes a system most capable of being a process in its own right?

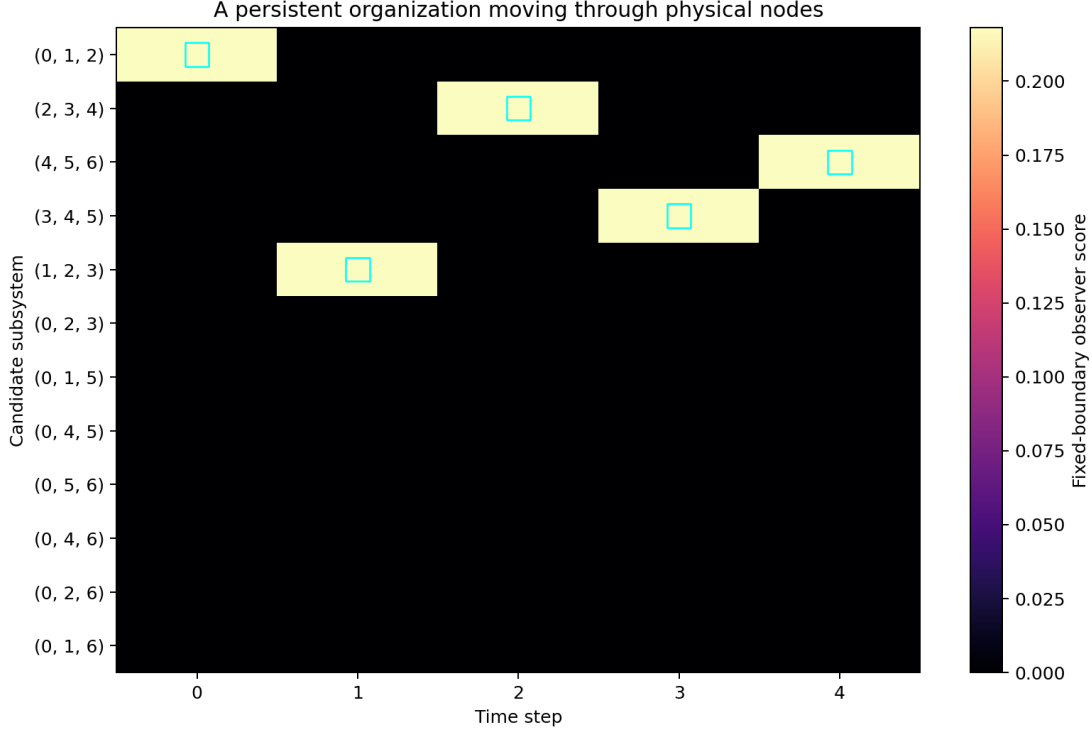


Figure 1: Fixed-boundary scores across time. Cyan squares mark the world-tube selected by the path action. This is a proof-of-concept recovery experiment, not evidence about biological consciousness.

## References

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